

Part I — Analytical Derivation

Q1. What does Θ_0 represent?

$\Theta_0 = \{\theta_t = \nabla\phi(X_t) : \phi \in \mathcal{D}\}$ is the set of *functionally generated trading strategies* from Fernholz’s SPT. Portfolio weights depend on X_t only via $\nabla\phi$, giving growth rates expressible in terms of ϕ and the ergodic density p alone, without reference to the drift of X .

Q2. Derive $g(\theta_0, P)$ via Itô’s formula.

Since $\log V_T^\theta = \int_0^T \theta_t^\top dX_t - \frac{1}{2} \int_0^T \theta_t^\top c \theta_t dt$ and Itô on $\phi(X_t)$ gives $\int_0^T (\nabla\phi)^\top dX_t = \phi(X_T) - \phi(X_0) - \frac{1}{2} \int_0^T \text{Tr}(c\nabla^2\phi) dt$, substituting and dividing by T :

$$g(\theta_0; P) = \int_E \left[\frac{1}{2} \text{Tr}(c\nabla^2\phi) - \frac{1}{2} (\nabla\phi)^\top c (\nabla\phi) \right] p(x) dx.$$

Boundary terms vanish by ergodicity. Since g depends on ϕ, c, p only (not on the drift, hence not on $P \in \Pi$), Θ_0 has **growth-rate invariance**, and the robust problem reduces to $\sup_{\phi \in \mathcal{D}} \int_E \left[\frac{1}{2} \text{Tr}(c\nabla^2\phi) - \frac{1}{2} |\nabla\phi|_c^2 \right] p dx$.

Q3. Integration by parts to eliminate the Hessian.

Setting $A(x) = cp(x)$ and using $\text{div } A = c\nabla p = -c\Sigma^{-1}x$ (Gaussian p), integration by parts on $\int \text{Tr}(A\nabla^2 e^\phi) dx$ yields $I := \int \frac{\text{Tr}(A\nabla^2 e^\phi)}{e^\phi} dx = \int (\nabla\phi)^\top c\Sigma^{-1}x p dx$. The Hessian vanishes, giving:

$$g(\phi) = \frac{1}{2} \int (\nabla\phi)^\top c\Sigma^{-1}x p dx - \frac{1}{2} \int (\nabla\phi)^\top c (\nabla\phi) p dx.$$

Q4. Compute $A^{-1} \text{div } A$.

$\text{div } A = c\nabla p = -c\Sigma^{-1}x p(x)$, so $A^{-1} \text{div } A = -\Sigma^{-1}x$.

Q5. LQ optimisation and solution for ϕ .

Quadratic ansatz $\phi(x) = \frac{1}{2}x^\top Mx$ gives $\nabla\phi = Mx$, $\nabla^2\phi = M$, and using $\int xx^\top p dx = \Sigma$:

$$g(M) = \frac{1}{2} \text{Tr}(cM) - \frac{1}{2} \text{Tr}(cM\Sigma M).$$

Setting $dg/dM = \frac{1}{2}c - cM\Sigma = 0$ gives $M^* = \frac{1}{2}\Sigma^{-1}$, hence $\phi^*(x) = \frac{1}{4}x^\top \Sigma^{-1}x$.

Q6-Q7. Robust-optimal strategy and growth rate.

$$\theta^*(x) = \nabla\phi^*(x) = \frac{1}{2}\Sigma^{-1}x, \quad g^* = \frac{1}{8} \text{Tr}(c\Sigma^{-1}) = \frac{1.0330}{8} = 0.1291.$$

θ^* is automatically robust since g is invariant over Π for all $\theta \in \Theta_0$. The growth rate depends only on c and Σ , not on drift B , immune to model uncertainty.

Part II — Adversarial Training

Q1–Q2. OU dynamics and Fokker–Planck constraint.

The d -dimensional OU process $dX_t = -BX_t dt + \sigma dW_t$ with $c = \sigma\sigma^\top$ fixed. The Adversary perturbs B . For $X_t \sim \mathcal{N}(0, \Sigma)$ at stationarity, the Lyapunov equation requires $B\Sigma + \Sigma B^\top = c$.

Q3. Degrees of freedom.

$B \in \mathbb{R}^{d \times d}$ has d^2 entries; the Lyapunov equation gives $d(d+1)/2$ constraints, leaving $d(d-1)/2$ free parameters. For $d=2$: **1 degree of freedom** (scalar α parametrising the skew-symmetric part of B).

Q4. Neural network design and loss functions.

Agent: 3-layer MLP ($d \rightarrow 64 \rightarrow 64 \rightarrow 1$), Softplus activations, outputting $\phi(x)$; $\theta = \nabla\phi$ via autograd. **Adversary:** $B = B_{\text{sym}} + S$ with $B_{\text{sym}} = \frac{1}{2}c\Sigma^{-1}$ fixed and $S\Sigma$ skew-symmetric (1 learnable α), hard-coding the Fokker–Planck constraint. **Minimax game:** Agent maximises / Adversary minimises $L = \mathbb{E}_p[\frac{1}{2} \text{Tr}(c\nabla^2\phi) - \frac{1}{2} |\nabla\phi|_c^2]$.

Q5. Training procedure and stability.

Alternating gradient descent–ascent: (1) 2 adversary steps on α ; (2) 3 agent steps on ϕ -weights. Growth rate estimated via Monte Carlo on OU paths (with burn-in). Stability: hard-coded constraint, Adam (agent 3×10^{-3} , adversary 10^{-2}), asymmetric step counts, initialisation from $\mathcal{N}(0, \Sigma)$.

Q6. Results.

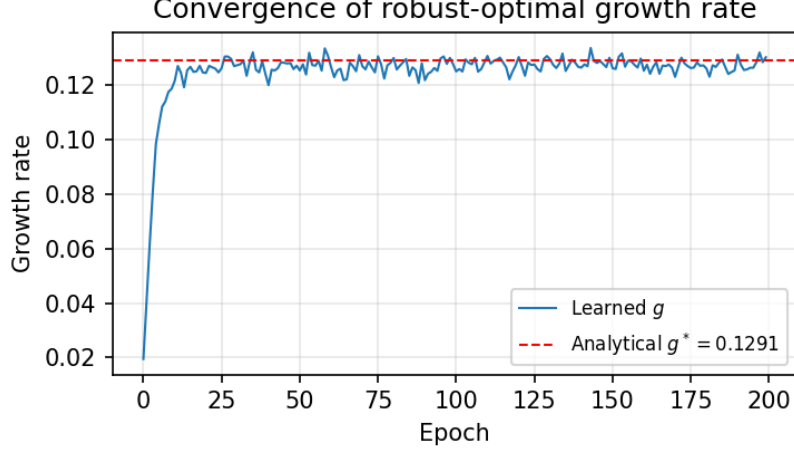


Figure 1: Learned growth rate converges to $g \approx 0.1303$; analytical $g^* = 0.1291$ (rel. error $< 1\%$).

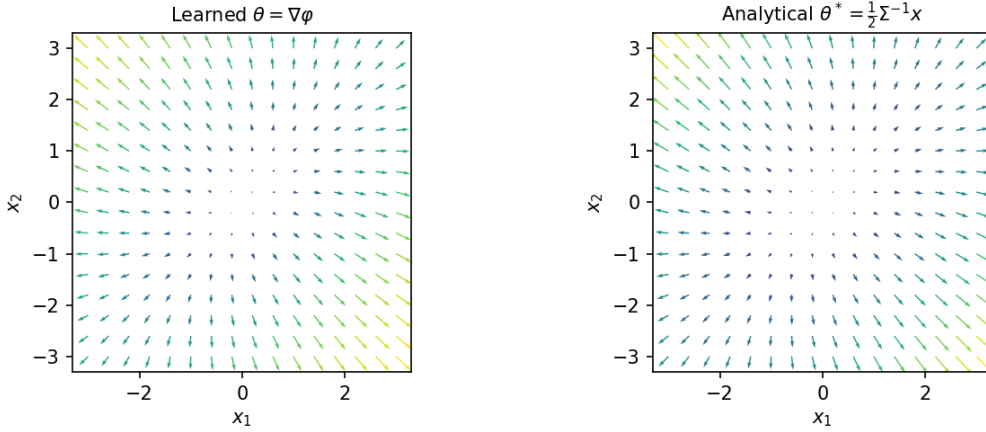


Figure 2: Learned $\theta = \nabla \phi$ (left) vs. analytical $\theta^* = \frac{1}{2} \Sigma^{-1} x$ (right): visually indistinguishable linear inward-pointing fields. Adversary settles at $\alpha = 0$, confirming growth-rate invariance.

Q7. Comparison and discussion.

	Analytical	ML
Growth rate g^*	0.1291	0.1303 ($< 1\%$ error)
Strategy θ^*	$\frac{1}{2} \Sigma^{-1} x$	$\approx \frac{1}{2} \Sigma^{-1} x$ (learned)
Drift perturbation	invariant	$\alpha = 0$ (none found)

The ML results confirm all analytical predictions: (1) growth rate matches to within MC noise; (2) the learned strategy approximates the linear θ^* ; (3) the adversary finds no beneficial drift perturbation, numerically verifying growth-rate invariance of Θ_0 over Π . This validates [?]: $\phi^*(x) = \frac{1}{4} x^\top \Sigma^{-1} x$ is robust-optimal and immune to drift uncertainty.